Table 1: Residual-objective ablation under the Double Sphere model. All variants use identical initialization, train–test splits, and frozen holdout observations. Hybrid attains the lowest reprojection error on two of three sequences, while Spherical gives the best macro-average and high-angle accuracy.

(a) Held-out reprojection RMSE [px] ↓

Residual	Seq. A	Seq. B	Seq. C	Mean
Pixel	0.51933	0.50425	0.50084	0.50814
Spherical	0.51956	**0.50312**	0.50049	**0.50772**
Hybrid	**0.51926**	0.50373	**0.50026**	0.50775
Kalibr (DS)	0.62837	0.56662	0.61958	0.60486

(b) Angular accuracy and geometric stability

Residual	0–30°	30–50°	≥ 50°	Ray RMS
	Angular RMSE [mrad] ↓			[deg] ↓
Pixel	0.3317	0.3339	0.3859	**0.3024**
Spherical	0.3314	**0.3335**	**0.3853**	0.3082
Hybrid	**0.3312**	0.3336	0.3859	0.3103

Protocol. Seq. A/B/C denote 1444190-clear, 134853-clear, and 192347-clear, respectively. Results use a 70/30 split with seed 1337. Polar bins are assigned using the shared initialization camera. Ray RMS is the pairwise cross-sequence ray-curve disagreement over 1,422 image locations. Best results are in bold.